

ATIF KHAN

Embedded Hardware Design Engineer | High-Speed & Multilayer PCB Design | Power Electronics

Islamabad, Pakistan | +92 345 6888102 | atifdev44@gmail.com | [GitHub Engineering Portfolio](#)

PROFESSIONAL SUMMARY

Embedded Hardware Design Engineer with 4+ years of experience spanning schematic design, high-speed multilayer PCB layout, and system-level hardware integration for embedded, industrial, IoT, wearable, and consumer electronics products. Proven expertise in power electronics architecture, DC-DC conversion, battery management systems (BMS), and power-path switching, alongside RF front-end design and ATEX/IECEX-compliant hardware for hazardous-area applications. Proficient in Altium Designer, OrCAD, and KiCad for high-speed, impedance-controlled, multilayer board design, with hands-on experience in schematic capture, layout review, BOM generation, MCU bring-up, hardware-in-the-loop (HIL) validation, DFM/DFT compliance, and EMI/EMC mitigation. Experienced collaborating with cross-functional firmware, mechanical, and testing teams and coordinating directly with PCB fabricators and assembly vendors to deliver manufacturable, production-ready hardware. Holds a Bachelor of Science in Engineering Sciences (Optoelectronics) from GIKI. Skilled at translating client and product requirements into reliable, well-documented hardware from concept through certification.

CORE COMPETENCIES

- **PCB Design & Layout:** 2–8 layer, multilayer, impedance-controlled, high-speed, mixed-signal and RF boards
- **System-Level Hardware Architecture:** cross-functional collaboration with firmware, mechanical, and testing teams; system-level integration of embedded, analog, and digital subsystems
- **Power Electronics:** Buck/Boost/Buck-Boost converters, LDOs, Li-Ion/LiPo battery management (BMS), fuel gauging, power-path switching, reverse-polarity & ESD/TVS protection
- **Embedded Systems:** ARM Cortex-M (STM32, nRF52840), ESP32/ESP32-S3/C6, AVR, 8051, PIC, MCU bring-up and firmware integration
- **Compliance & Standards:** ATEX/IECEX hazardous-area design, IEC 60079-7:2013, DFM/DFT compliance, QA & compliance verification for Zone 1/Zone 2 deployments
- **RF & Wireless:** Antenna matching, LTE/NB-IoT, BLE 5.0, Wi-Fi 6, GNSS, LoRa, microstrip impedance-matched RF traces
- **Signal Integrity & EMI/EMC:** high-speed routing, thermal modelling, power integrity optimization, noise mitigation
- **Validation & Testing:** Hardware-in-the-Loop (HIL) simulation, bench testing with oscilloscopes, logic analyzers, function generators and power supplies, fault isolation
- **Program Delivery:** BOM management, DFM/DFT, component sourcing, PCB fabricator & assembly vendor coordination, hardware documentation & version control, cross-functional project management

TECHNICAL SKILLS

- **EDA / CAD Tools:** Altium Designer, OrCAD, KiCad 9.0, EasyEDA Pro, LTspice, Proteus, SolidWorks, FreeCAD
- **Microcontrollers & SoCs:** STM32, ESP32 / ESP32-S3 / ESP32-C6, Nordic nRF52840, AVR (ATtiny, ATmega), AT89S52 (8051), PIC, Renesas Synergy, MAX78000 edge-AI SoC
- **Communication Protocols & Interfaces:** SPI, I2C, UART, CAN, RS-232, RS-485, USB, Ethernet, JTAG/SWD, UPDI
- **Wireless / IoT:** Quectel & SIMCom LTE-M/NB-IoT modules, BLE 5.0, Wi-Fi 6, GNSS, NFC, LoRa
- **Lab Instrumentation:** oscilloscopes, logic analyzers, function generators, bench power supplies
- **Programming & Simulation:** C, C++, Python, Embedded C, MATLAB, Simulink
- **Other:** Microsoft Office, WordPress, HTML/CSS, technical documentation & version control, project management

PROFESSIONAL EXPERIENCE

Embedded Designer

Jan 2026 – Present

Embedded Hardware & Power Electronics Design | Karachi, Pakistan ([embeddeddesigner.com](#))

- Design and develop embedded hardware and power management circuits, covering schematic capture, high-speed multilayer PCB layout, layout reviews, and prototype validation for client products.
- Architect power delivery subsystems including DC-DC conversion, battery charging, and power-path switching for embedded and IoT-enabled devices.
- Create schematics and generate BOMs using Altium Designer and OrCAD, and coordinate directly with PCB fabricators and assembly vendors to ensure DFM/DFT compliance.

- Collaborate with cross-functional firmware, mechanical, and testing teams on system-level hardware architecture, from concept through prototyping and validation.
- Select components and define design-for-manufacturability (DFM) and design-for-testability (DFT) requirements in collaboration with cross-functional engineering teams.
- Perform hardware bring-up, debugging, and validation testing using oscilloscopes, logic analyzers, and bench power supplies to ensure design reliability and compliance with project specifications.
- Maintain hardware documentation, revision/version control, and project timelines across concurrent client engagements.

Hardware Design Engineer

Jan 2025 – Jul 2025

Smart Computings Pvt Ltd | Lahore, Pakistan (www.smartcomputings.com)

- Led schematic verification and PCB-level design analysis for an STM32F407-based system integrating 4G LTE connectivity, relay control, SoC switching between primary and secondary power sources, and 3-phase AC monitoring.
- Conducted extensive lab prototyping and testing of digital I/O circuits using oscilloscopes, logic analyzers, and function generators for signal verification and fault isolation.
- Validated peripheral interfaces (SPI, UART, I2C) and industrial communication protocols including RS-232, RS-485, CAN, USB, and Ethernet.
- Performed board bring-up and power-domain verification for a Renesas Synergy MCU, ensuring correct voltage sequencing and regulator configuration.
- Integrated SIMCom and Quectel cellular modules for remote telemetry and validated pressure/temperature sensor interfaces.
- Simulated Hardware-in-the-Loop (HIL) conditions and proposed circuit enhancements, including reverse-polarity protection, flyback diode optimization, and ESD/TVS placement strategies.
- Reverse-engineered European ATEX certified EVC hardware to localize components and reduce cost while preserving safety and functional integrity.
- Designed and tested industrial power regulation modules (24V to 5V/3.3V) in accordance with IEC 60079-7:2013 for explosive atmospheres.
- Analysed schematic and design compliance for Zone 1 and Zone 2 ATEX gas-flow computers, ensuring critical safety and environmental standards were met.
- Led QA and compliance verification for field deployment in hazardous (Zone 2) locations, including final assembly checks.
- Managed firmware flashing, device initialization, and complete hardware validation of ATEX-compliant systems.
- Used Altium Designer for complex multilayer, impedance-controlled, high-speed PCB layout, thermal modelling, EMI mitigation, and power integrity optimization.
- Oversaw hardware QA, BOM verification, hardware documentation, and delivery of production-ready hardware modules for embedded field deployments.

R&D Engineer

Jun 2024 – Dec 2024

NR Trading Pvt Ltd. | Taxila, Pakistan

- Designed multilayer and RF PCBs, handling component selection and design optimization.
- Developed embedded systems using AT89S52, STM32, and PIC microcontrollers, including schematic design and simulation.
- Reverse-engineered existing PCBs and delivered enhanced redesigns with improved performance.
- Managed prototyping, testing, and troubleshooting to ensure high-quality, on-time project delivery.

Embedded Systems Firmware Developer

Aug 2023 – Sep 2023

Chip Soul Technology Pvt Ltd | Peshawar, Pakistan

- Gained hands-on experience in embedded systems development and C programming for the 8051 microcontroller family.
- Worked on microcontroller interfacing and control-system projects.

Hardware Design Engineer

Jul 2022 – Sep 2022

Chip Soul Technology Pvt Ltd | Peshawar, Pakistan

- Learned detailed PCB design practices, including multilayer board development.
- Gained experience in component selection for optimized circuit performance.
- Assisted in layout and simulation of electronic circuits across multiple projects.

Embedded Hardware Design Engineer

Dec 2021 – Present

Remote / Online

- Deliver end-to-end embedded hardware solutions for global clients, spanning schematic design, high-speed multilayer PCB layout, component selection, and hardware bring-up for embedded, IoT, and power electronics systems.

- Collaborate with clients from concept through prototyping and validation, ensuring designs meet performance, reliability, and manufacturability requirements in line with industry best practices.

KEY PROJECTS

AI-Enabled Electrical Fault Detector — Edge-AI Sensor Node

- Designed a 6-layer, hierarchical mixed-signal PCB (KiCad 9.0) for an edge-AI arc-fault detection sensor node powered by the MAX78000 (Arm Cortex-M4F + CNN hardware accelerator) for on-device, real-time fault classification.
- Built an arc-fault analog front-end based on TI's TIDA-01385/TIDA-010231 reference design with CT sensing, active filtering, and a 12-bit SAR ADC; implemented CT-based energy harvesting and a 4-rail (1.2V/1.8V/2.5V/3.3V) power architecture.
- Enabled dual-radio IoT connectivity (LTE-M/NB-IoT + Wi-Fi 6/BLE) with multi-modal sensor fusion (IMU, environmental, acoustic) and full Li-Ion battery management with fuel gauging.

BLE Motion & Biometric Training Device — Wearable Sensor

- Designed a compact 4-layer wearable PCB (Altium Designer) around the Nordic nRF52840 (BLE 5 + NFC tap-to-pair) with a 6-axis IMU for motion, posture, and cadence tracking.
- Engineered complete power architecture: Li-Po charging with Model Gauge fuel gauging, buck-boost regulation, and a pushbutton power controller for clean on/off and kill-switch operation.
- Implemented USB-C charging/programming, capacitive touch HMI, and on-board NAND flash logging for offline-capable training sessions.

Smart Liquid Dispensing & Motor Control Device (KIPFLO)

- Designed a 4-layer PCB (KiCad 9.0) built around the ESP32-S3, integrating a dual H-bridge motor driver, Hall-effect position sensing, capacitive liquid-level sensing, and a 3-axis accelerometer for orientation/fall detection.
- Implemented a complete battery charging system with Dynamic Power Path Management (DPPM) and a POGO-pin charging dock with automatic power-path switching for safe charging.

Multi-Rail Power Distribution Unit (12V, 9A)

- Designed a 2-layer power distribution PCB (Altium Designer) supporting three independently fused 12V output rails (9A/3A/5A) for industrial and automotive applications.
- Implemented ideal-diode reverse-polarity protection, dual TVS surge protection, and high-current vibration-tolerant connectors for robust field deployment.
- Additional projects — including a cellular IoT GNSS tracker, an AI-powered OCR text-recognition device, and a low-power wearable digital counter — are detailed in the [GitHub portfolio](#).

EDUCATION

B.S. in Engineering Sciences — Optoelectronics

Sep 2020 – Jun 2024

Ghulam Ishaq Khan Institute of Engineering Sciences & Technology (GIKI), Swabi, Pakistan

- Final Year Project: NOMA-based Visible Light Communication (VLC) system for two users covering modulation, user separation, interference analysis, and system performance evaluation.